

Day 99: Machine Learning Fundamentals

1. Day Number

Day 99

2. Topic Name

Machine Learning, Features, Labels, and Workflow

3. Connection

You now understand basic **data handling** and some of the mathematics used in machine learning, such as statistics, probability, derivatives, and gradients.

Today, you will connect those ideas and learn what **machine learning actually does**.

The basic idea is:

Data → Learn patterns → Make predictions

4. Important Topics

Today you should understand these terms:

- **Machine learning (ML):** teaching a computer to learn patterns from data instead of giving it every rule manually.
 - **Dataset:** a collection of data used for analysis or machine learning.
 - **Example / observation:** one individual row or case in a dataset.
 - **Feature:** information used as input to a model.
 - **Label / target:** the value we want the model to predict.
 - **Model:** a mathematical system that learns relationships from data.
 - **Prediction:** the model's estimated output for new input data.
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5. Foundational Notes

Imagine this dataset:

Size (sq ft)	Bedrooms	Age	Price
1000	2	10	50
1500	3	5	75
2000	4	2	100

Each **row** represents one house.

Therefore, each row is one **observation** or **example**.

Suppose we want to predict house prices.

Then:

Features:

Size, Bedrooms, Age

These contain information that may help predict the price.

Label / Target:

Price

This is what we want the model to learn to predict.

Conceptually:

```
Features
  ↓
Machine Learning Model
  ↓
Prediction
```

For example:

```
Size = 1800
Bedrooms = 3
Age = 4
  ↓
  Model
  ↓
Predicted Price
```

You do **not** need to know how the model calculates that prediction yet.

6. Traditional Rules vs Machine Learning

Traditional programming usually looks like:

```
Data + Rules → Answer
```

You tell the computer exactly what rules to follow.

For example:

```
if temperature > 30:
    print("Hot")
else:
    print("Not hot")
```

You created the rule:

```
temperature > 30
```

Machine learning is different.

Instead of manually creating every rule, we provide examples:

```
Data + Correct Answers → Learning Algorithm
                        ↓
                        Model
```

The model tries to discover useful patterns.

For example, instead of writing:

```
If house size is X
and bedrooms are Y
and age is Z,
then price should be ...
```

we could provide many previous houses with their actual prices.

The machine-learning system tries to learn the relationship between:

```
Size
Bedrooms
Age
```

and:

```
Price
```

A useful simplification is:

Traditional programming: humans write the rules.

Machine learning: the computer learns useful patterns from examples.

7. Supervised vs Unsupervised Learning

There are many types of machine learning. For now, focus on two broad categories.

Supervised Learning

In supervised learning, the dataset contains the **correct target or label**.

Example:

Size	Bedrooms	Price
1000	2	50
1500	3	75
2000	4	100

If we want to predict Price, we already have known prices in our training examples.

The model can learn:

Features → Label

Common supervised-learning tasks include predicting things such as:

House information → Price
 Customer information → Will purchase or not
 Email information → Spam or not spam

Unsupervised Learning

In unsupervised learning, there is usually **no target label telling the model the correct answer**.

Instead, we ask the system to discover patterns or groups in the data.

For example:

Customer age
 Customer spending
 Number of purchases

We might ask:

| Are there groups of customers with similar behavior?

The system could discover groups such as:

Group A → low spending
 Group B → medium spending
 Group C → high spending

At this stage, remember:

Supervised
 → Features + known label

Unsupervised
 → Features, but usually no known label

8. Easy Example: Predicting House Price

Suppose we have:

House	Size	Bedrooms	Age	Price
A	900	2	15	45
B	1400	3	8	70
C	1800	3	4	90

Our goal is:

| Predict the price of a house.

One observation might be:

House B

Its features are:

Size = 1400
Bedrooms = 3
Age = 8

The value we want to predict is:

Price

So conceptually:

[1400, 3, 8] → Model → Price prediction

A dataset containing many such examples could eventually be used to train a model.

Today, however, we are **only identifying the parts of the ML problem**. We are not training anything yet.

9. Problem Statement

Consider this small house dataset:

House	Size	Bedrooms	Age	Price
A	1200	2	10	60
B	1600	3	6	80
C	2100	4	2	110

Your goal is to identify:

1. What represents an **observation/example**?
2. Which columns could be **features**?
3. Which column should be the **label/target** if the goal is to predict house price?
4. How many observations are present?
5. What information would you give to a future model to predict the price of a new house?

Do not train a model.

10. Concepts Used

This exercise uses:

- datasets
- rows and columns
- observations
- features
- labels
- predictions
- supervised learning
- basic data preparation ideas

Notice how this connects directly to your earlier Pandas lessons.

A DataFrame might eventually contain:

```
rows → observations
columns → variables
selected input columns → features
prediction column → target
```

11. Thought Process

When looking at a machine-learning problem, first ask:

What is one example?

Usually one row represents one example.

Then ask:

What information do I already know about each example?

Those values may become features.

Next ask:

What am I trying to predict?

That usually becomes the target.

For the house example, think:

```
What information describes the house?  
    ↓  
Possible features  
  
What value do I want the model to estimate?  
    ↓  
Target
```

This simple way of thinking will help with many future ML problems.

12. Beginner-Friendly ML Workflow

A simplified machine-learning workflow looks like this:

```
1. Define the problem  
    ↓  
2. Collect data  
    ↓  
3. Inspect the data  
    ↓  
4. Clean the data  
    ↓  
5. Choose features and target  
    ↓  
6. Split data for training/testing  
    ↓  
7. Train a model  
    ↓  
8. Evaluate the model  
    ↓  
9. Make predictions
```

Several of these steps should already look familiar.

You previously learned about:

CSV / JSON / databases



Pandas



Data inspection



Cleaning



Encoding / scaling



Features and labels

You are now reaching the point where these data-processing skills connect to actual machine-learning models.

For **Day 99**, focus mainly on steps 1–5.

We will not train a model yet.

13. Easy Edge Cases

Missing Feature

Suppose a new house contains:

Size = 1500

Bedrooms = ?

Age = 5

A feature is missing.

This can create problems because the model may expect all required input values.

Possible future solutions include:

fill the missing value

remove incomplete data

use a model that can handle missing values

For now, simply remember:

Check whether required features are missing.

Unavailable Label

Suppose we have:

Size	Bedrooms	Age	Price
1500	3	5	?

If we are trying to learn how to predict price, the missing price means this example does not provide a known target value.

However, this could also represent a **new house** for which we actually want the trained model to make a prediction.

The meaning depends on the stage of the ML workflow.

14. Expected Identification

When solving today's exercise, your identification should have this general form:

Observation:
One _____ in the dataset

Features:
Columns that describe _____

Label:
The value that we want to _____

Number of observations:
Count the number of _____

You should also be able to describe a future prediction conceptually:

New house features
↓
Model
↓
Predicted house price

The important skill today is not calculation. It is correctly identifying the **role each part of the data plays**.

15. Hint Only

Look at the question:

 | **What are we trying to predict?**

That column is likely your **target**.

Then look at the other useful information describing each house:

Size
Bedrooms
Age

Ask yourself whether those values could help predict the target.

Finally, remember:

| **One row usually represents one observation/example.**

Do not build or train a model yet. The goal of Day 99 is simply to understand how a normal dataset becomes a **machine-learning problem**.